



Mitochondrial Transport Defects as a Potential Biomarker in Patient-derived Cells

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Summary

We screened control and patient lines for changes in mitochondrial motility. Mitochondria motility was analyzed in both iPSCs and after differentiation to iNeurons. In iNeurons, motility was assayed both in standard culture conditions and after the addition of mitochondrial stressors. In total, 14 PPMI lines were used for this project (healthy controls (PPMI 3411, 3668, 3658, 3452, 3453, 3480, 3955, 3952), SNCA-A53T (PPMI 42378, 40760, 41430), LRRK2-G2019S (PPMI 40273), Sporadic PD (PPMI 3409, 3664)). From the G-banded karyotyping results, healthy control (PPMI 3952) showed a nonclonal finding by containing chromosomal aberrations, thus this line was excluded from the screening, whereas sporadic PD (PPMI 3664) showed abnormal karyotype by showing an extra copy of chromosome 21 (Trisomy 21). Despite the result, PPMI 3664 was included in the screening. **From the iPSCs**, one SNCA-A53T line (PPMI 41430) had less mitochondrial motility than the controls, but the others did not and so this is not attributable to the SNCA mutation. In the two sporadic PD lines (PPMI 3409 and 3664), mitochondrial motility was increased relative to the control lines. **From the iNeurons**, the SNCA-A53T lines (PPMI 41430) again showed decreased mitochondrial motility relative to the control lines. In one of the sporadic PD lines (PPMI 3409) mitochondrial motility was higher than in control lines. The mitochondrial stressors (antimycinA, oligomycin, and rotenone) depressed motility in all the lines without evoking a clear difference in the responses among them. More screening is required to find out the significance of the changes in sporadic PD line.





Method

The 14 iPSCs obtained from PPMI were tested for mycoplasma and G-banded karyotyping. PPMI 3952 was excluded due to an abnormal non-clonal karyotype. To analyze movement in iPSCs, iPSCs were transduced with pLV-EF1 alpha-mito-dsRed and live-imaged for 3 min using Nikon TiEclipse 40X and 60X objective. To quantify motility in the iPSCs, where motility is not linear and mitochondria can be in a reticulum rather than discrete organelles, we used the QuoVadoPro software package (2). This software uses the per pixel variance of fluorescent intensity as a proxy for organelle motility. To analyze motility in iNeurons, iPSCs were differentiated into cortical-like neurons using a modification of the protocol of Zhang et al. (1). iPSCs were differentiated into neuronal cells by neurogenin-2 (Ngn2/puromycin resistance) transduction. To label mitochondria, pLV-EF1 alpha-mito-dsRed was transduced in parallel with Ngn2 on the same days. Puromycin selection begins on day 1, addition of AraC begins on day2 to decrease non-neuronal cell proliferation and then 20,000 cortical neurons/well were plated in 96-well plate on day 3. On day14 post-differentiation, mitochondria were imaged by means of a time-lapse, high-throughput platform called Arrayscan XTi (ThermoFisher) using a 20X objective with 2x2 binning (3). For each well in a 96-well plate, 8 fields were imaged. Movies were collected at 30 frames per field at the maximum speed of 2 Hz. For image analysis of iNeuron movies, we used a previously published protocol that distinguishes stationary from moving mitochondria and tracks their movements (3). For the image analysis, fields containing undifferentiated stem cells in addition to the iNeurons, or fields that were out of focus were excluded from analysis. We report the “Integrated Distance” parameter for each line or condition, a measure of the average distance traveled by mitochondria in that field (3). To compare mitochondria motility in healthy controls and PD lines, 4 healthy controls and 3 PD lines were differentiated in parallel on the same days and plated on the same 96-well plate. For mitochondria motility analysis in the stress condition, DMSO control was always tested with stressor on the same plate. At least 3 wells were averaged for each line and condition. For the stress condition, iNeurons were incubated in 25nM Antimycin A, 25nM Oligomycin, 50nM Rotenone for 1h prior to imaging.

Statistical analysis was done in Graphpad Prism, Microsoft Excel, and MATLAB. **<0.01, ****<0.0001. For statistical analysis of integrated distance, the nonparametric comparison called Kolmogorov-Smirnov (KS) statistic were used. The test value here are presented as mean $\pm$ SEM.

References

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